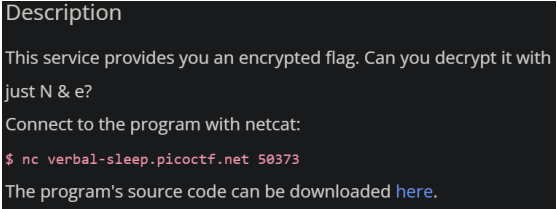


Easy – EVEN RSA CAN BE BROKEN???

A. The Case

Dalam kasus ini, kita diberi sebuah instance yang harus dirun. Menggunakan netcat, kita akan connect ke browser yang diberikan.

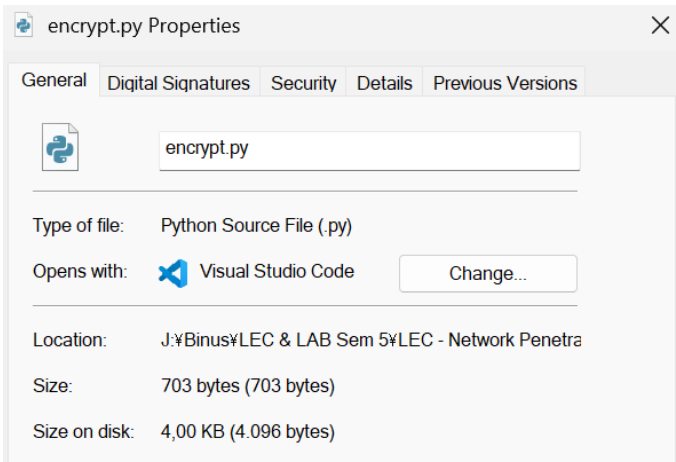


Menjalankan command tersebut, kita akan diberi N, e, dan cyphertext:

```
LAPTOP-9V3V240C:/mnt/host/j/Binus/LEC & LAB Sem 5/LEC - Network Penetration Testing/Assignment#4# nc verbal-sleep.picoc
f.net 50373
N: 215669428957828084926935186832824211011060271916482533478780102697513827746259591116088802989195690680827242302344702
68012189441898841830778111942756409158
e: 65537
cyphertext: 191237638255959977740489014031938475040708428672453736510881151042256579779914958449200871483547545229218452
46602757138762909972141952421071917886464070987
```

- N:
21566942895782808492693518683282421101106027191648253347878010269751382774625959111608880298919569068082724230234470268012189441898841830778111942756409158
- e: 65537
- cyphertext:
19123763825595997774048901403193847504070842867245373651088115104225657977991495844920087148354754522921845246602757138762909972141952421071917886464070987

Kita juga diberikan source code dari program tersebut:



```
encrypt.py 2 X
J: > Binus > LEC & LAB Sem 5 > LEC - Network Penetration Testing > Assignment#4 > encrypt.py X

1 from sys import exit
2 from Crypto.Util.number import bytes_to_long, inverse
3 from setup import get_primes
4
5 e = 65537
6
7 def gen_key(k):
8     """
9     Generates RSA key with k bits
10    """
11    p,q = get_primes(k//2)
12    N = p*q
13    d = inverse(e, (p-1)*(q-1))
14
15    return ((N,e), d)
16
17 def encrypt(pubkey, m):
18     N,e = pubkey
19     return pow(bytes_to_long(m.encode('utf-8')), e, N)
20
21 def main(flag):
22     pubkey, _privkey = gen_key(1024)
23     encrypted = encrypt(pubkey, flag)
24     return (pubkey[0], encrypted)
25
26 if __name__ == "__main__":
27     flag = open('flag.txt', 'r').read()
28     flag = flag.strip()
29     N, cypher = main(flag)
30     print("N:", N)
31     print("e:", e)
32     print("cyphertext:", cypher)
33     exit()
```

B. Finding the Flag

Melalui source code yang diberikan, kita bisa melihat di dalam def gen_key(k) bahwa value N berasal dari perkalian bilangan prima p dan q. Value N seharusnya ganjil, tetapi di kasus ini valuenya genap yang menyebabkan vulnerability (salah satu bilangan prima menjadi 2 dan yang kedua menjadi N/2)

```
p,q = get_primes(k//2)

N = p*q
```

2156694289578280849269351868328242110110602719164825334787801026975138277462595911160888029891! Factorize!

Result:

status (?)	digits	number
FF	155 (show)	2156694289...58<155> = 2 · 1078347144...79<155>

- p = 2 → karena N genap
- q =
10783471447891404246346759341641210550553013595824126673939005134875691387312979555804440149459784534041362115117235134006094720949420915389055971378204579

Kemudian di dalam perhitungan value d, kita bisa mengubah (p-1)*(q-1) menjadi (q-1) karena value p adalah 2 (dimana 2-1 menjadi 1, dan value yang dikali 1 akan menghasilkan value yang sama).

```
d = inverse(e, (p-1)*(q-1))
```

Dengan menggunakan kode ini, kita bisa menghitung value d:

- d =
2040956709029174027979388786085990748265248342807917165319429020141264872792929160782432452109482694664831403267073326551590687987802570067074328531596649

```
p = 2
q = N//p
phi_N = (p-1) * (q-1)
d = pow(e, -1, phi_N)
print(d)
```

```
PS J:\Binus\LEC & LAB Sem 5\LEC - Network Penetration Testing\Assignment#4> & C:/Users/Just1ce/AppData/Local/Programs/Python/Python312/python.exe "j:/Binus/LEC & LAB Sem 5/LEC - Network Penetration Testing/Assignment#4/decrypt.py"
2040956709029174027979388786085990748265248342807917165319429020141264872792929160782432452109482694664831403267073326551590687987802570067074328531596649
```

Setelah mendapatkan value d, kita bisa mendecrypt cyphertext tersebut melalui long_to_bytes (kebalikan dari bytes_to_long di function encrypt).

- Flag: picoCTF{tw0_1\$_pr!m37dbe6984}

```
def encrypt(pubkey, m):
    N,e = pubkey
    return pow(bytes_to_long(m.encode('utf-8')), e, N)
```

```
cyphertext = 1912376382559599777

m = pow(cyphertext, d, N)

flag = long_to_bytes(m)
print(flag)
```

```
PS J:\Binus\LEC & LAB Sem 5\LEC - Network Penetration Testing\Assignment#4> & C:/Users/Just1ce/AppData/Local/Programs/Python/Python312/python.exe "j:/Binus/LEC & LAB Sem 5/LEC - Network Penetration Testing/Assignment#4/decrypt.py"
2040956709029174027979388786085990748265248342807917165319429020141264872792929160782432452109482694664831403267073326551590687987802570067074328531596649
b'picoCTF{tw0_1$_pr!m37dbe6984}'
```

Setelah kita masukkan flag tersebut, ternyata flag yang kita dapatkan sudah benar.

EVEN RSA CAN BE BROKEN???

Easy

Cryptography

picoCTF-2025

browser_webshell_solvable

AUTHOR: MICHAEL CROTTY

Description

This service provides you an encrypted flag. Can you decrypt it with just N & e?

Connect to the program with netcat:

\$ nc verbal-sleep.picoctf.net 64135

The program's source code can be downloaded [here](#).

debug info: [u889823 e: p: c470 L296802]

13,541 users solved

🔖

picoCTF{tw0_1\$_pr1m37dbe6984}

Submit Flag

👤

×

This challenge launches an instance on demand.

Its current status is: **RUNNING**

Instance Time Remaining: 14:53

Restart Instance

Hints

1

2

3

👍

96% Liked

👍

Cryptography

👤

✓

Easy

EVEN RSA CAN BE BROKEN???

13,542 solves

96%

👍